KANISHQ SINGHAL

Software Engineering | Computer Science

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SUMMARY

Software Engineering undergraduate experienced in designing full-stack applications utilizing Java, Spring Boot, MongoDB, PostgreSQL, and React.js. Proven ability to build scalable REST APIs, integrate AI-powered features, and develop real-time collaborative applications supporting concurrent users via WebSockets. Published researcher at APSIPA ASC in audio deepfake detection, 600+ DSA problems solved across competitive programming platforms, 25+ contests, and a 1700+ LeetCode rating.

EDUCATION

Year	Degree/Certificate	Institute/School	CGPA/%
Jul 2027	B. Tech in Computer Science Engineering (CSE)	The LNM Institute of Information Technology, Jaipur	7.32
2023	Class XII	The Palace School, Jaipur	89%

TECHNICAL SKILLS

- **Languages:** Java, Python, C, SQL, HTML
- **Frameworks & Technologies:** Spring Boot, REST APIs, WebSockets
- **Databases & Tools:** MongoDB, PostgreSQL (Neon), MySQL, Git, GitHub
- **Core Concepts:** DBMS, Object-Oriented Programming, Operating Systems, Computer Networks

EXPERIENCES

AI/ML Intern | UpValue | Certificate **|** Report (Jun 2025 – Jul 2025)

- Engineered AI-powered counterfeit detection pipeline with database storage and carousels, processing metadata and 1,500+ images.
- Designed CNN and CLIP-based models on 1,200+ sneaker images, achieving 87% classification accuracy.
- Automated data preprocessing to optimize quality, validating performance via LPIPS, cosine similarity, and Jaccard metrics.
- Integrated object detection to localize products in user uploads, boosting prediction reliability for digital commerce.

Part-Time AI Model Developer | Outlier (Oct 2024 – Dec 2024)

- Drove AI model optimization by evaluating outputs against reference formats and executing comprehensive test suites.
- Delivered actionable feedback that enhanced formatting precision across 5+ model outputs, significantly reducing manual correction time.

PUBLICATIONS

Singhal, Kanishq. "**Quefrency Approach to Audio Deepfake Detection**." *Proceedings of the Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, Macau, China. (May 2024 – Aug 2024)

URL: https://ieeexplore.ieee.org/abstract/document/10848775

- Proposed quefrency-based representations for detecting audio deepfakes, conducting rigorous cepstral versus spectral analysis.
- Introduced a novel replayed-deepfake scenario via the rev-FoR-rerec dataset, mapping room-based reverberation topologies.
- Outperformed traditional mel-spectrograms in 3 deepfake conditions, demonstrating high robustness across ASVSpoof and FoR datasets.

PROJECTS

AI-Powered Collaborative Drawing Tool | GitHub **|** Report (Sept 2025 – Present)

- Architected a full-stack collaborative platform utilizing React.js, REST APIs and Spring Boot with PostgreSQL (Neon) and Prisma ORM for persistent canvas storage.
- Designed a backend pipeline integrating Gemini and OpenAI APIs to translate natural-language prompts into structured actions, reducing manual UI interactions by 70% and latency by 40%.
- Implemented WebSocket-based real-time collaboration for 20 concurrent users, achieving synchronized updates, secure JWT management, and 98.8% shape interpretation accuracy.

Delivery Dispatch Simulator — Rider Assignment & Route Planning Engine (Sept 2025 – Dec 2025)

- Built a Java and Spring Boot dispatch backend that assigns incoming orders across N riders on a 50×50 city grid.
- Implemented cost-based assignment over a min-heap weighing distance, load, and deadline, cutting late deliveries versus a greedy baseline.
- Designed A* routing with a time-window reservation table to keep concurrent riders off the same road segments.

POSITIONS OF RESPONSIBILITY

General Secretary | Cultural Council | LNMIIT (Aug 2025 – Present)

- Directed a 40+ member council to execute 50+ cultural events and flagship fests, engaging 15,000+ participants across partner institutes.
- Managed an annual fest and overall council budget exceeding INR 65 lakhs.

Team Lead | PR and Event Management | Vivacity (Sep 2023 – Mar 2026)

- Spearheaded PR and event management for 15+ flagship events, coordinating outreach across 20+ colleges and finalizing comprehensive expenditure reports for Vivacity ’26.
- Secured sponsorships worth INR 50,000+ and drove a 100% increase in campus participation during high-pressure festival executions.